



```
In [1]: a=10  
print(a)
```

10

this is python notebook

```
In [3]: # pip install numpy
```

```
In [4]: import numpy as np
```

```
In [9]: a=[1,2,3,4,5,6]  
b=[7,8,9,10,11,12]
```

```
In [10]: arr= np.array(a)  
arr2= np.array(b)
```

```
In [7]: arr
```

Out[7]: array([1, 2, 3, 4, 5, 6])

```
In [8]: print(arr)
```

[1 2 3 4 5 6]

```
In [11]: a+b
```

Out[11]: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

```
In [12]: arr+arr2
```

Out[12]: array([8, 10, 12, 14, 16, 18])

```
In [13]: arr*5
```

Out[13]: array([5, 10, 15, 20, 25, 30])

different methods of creating array

```
In [16]: np.arange(2,10,2)
```

Out[16]: array([2, 4, 6, 8])

```
In [17]: np.ones(5)
```

Out[17]: array([1., 1., 1., 1., 1.])

```
In [18]: np.zeros(5)
```

Out[18]: array([0., 0., 0., 0., 0.])

```
In [19]: np.eye(5)
```

```
Out[19]: array([[1., 0., 0., 0., 0.],
               [0., 1., 0., 0., 0.],
               [0., 0., 1., 0., 0.],
               [0., 0., 0., 1., 0.],
               [0., 0., 0., 0., 1.]])
```

```
In [44]: np.random.rand(20)
```

```
Out[44]: array([0.85653119, 0.18086864, 0.58245561, 0.01922881, 0.05233017,
               0.15183113, 0.37473763, 0.92783625, 0.89718266, 0.27028831,
               0.03012896, 0.24145201, 0.83439548, 0.70680003, 0.52051645,
               0.12499921, 0.18153543, 0.39013561, 0.50993078, 0.33833454])
```

```
In [59]: np.random.randint(7)
```

```
Out[59]: 2
```

```
In [60]: arr= np.arange(10,20)
```

```
In [61]: arr
```

```
Out[61]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [62]: arr.size
```

```
Out[62]: 10
```

```
In [63]: arr.shape
```

```
Out[63]: (10,)
```

```
In [64]: arr.ndim
```

```
Out[64]: 1
```

```
In [66]: arr3=np.eye(5)
```

```
In [67]: arr3
```

```
Out[67]: array([[1., 0., 0., 0., 0.],
               [0., 1., 0., 0., 0.],
               [0., 0., 1., 0., 0.],
               [0., 0., 0., 1., 0.],
               [0., 0., 0., 0., 1.]])
```

```
In [68]: arr3.ndim
```

```
Out[68]: 2
```

```
In [69]: a=[[1,2,3], [4,5,6]]
```

```
In [71]: np.array(a).ndim
```

```
Out[71]: 2
```

```
In [72]: arr
```

```
Out[72]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [73]: arr+100
```

```
Out[73]: array([110, 111, 112, 113, 114, 115, 116, 117, 118, 119])
```

```
In [74]: arr*5
```

```
Out[74]: array([50, 55, 60, 65, 70, 75, 80, 85, 90, 95])
```

```
In [75]: arr/3
```

```
Out[75]: array([3.33333333, 3.66666667, 4.          , 4.33333333, 4.66666667,
                5.          , 5.33333333, 5.66666667, 6.          , 6.33333333])
```

```
In [ ]:
```